

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular and Supplementary University Examination – Summer 2024 Course: B. Tech. Branch: Civil Engineering / Civil & Infra Semester: VI Subject Code & Name: Design of R C Structures & BTCVC 601 _Y22/ BTCVC601/BTCIC601 Max Marks: 60 Date:13/06/2024 Duration: 4 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of IS456-2000, IS875-205 & column interaction curves is permitted			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	a) Derive stress block parameters for M25 & Fe500 as per WSM	CO1	4
B)	What is ultimate load theory ? Explain in detail	CO1	4
C)	Explain working load, characteristic load, factored load, load factor & factor of safety.	CO1 & CO2	4
Q.2	Solve Any Two of the following.		
A)	Design a cantilever beam 3.0m effective span to carry point load 30 KN at free end & 20 KN point load at mid span. Supp. width is 300mm, Fck = 20MPa, Fy = 500 MPa, Sketch rein. details. Use WSM	CO2	4
B)	Find the longitudinal & shear reinforcement placed at 50mm effective cover in beam cross section 250x600mm. Fck = 20MPa, Fy = 500 MPa. Use WSM Sketch rein. details.	CO2	4
C)	Find the design axial load & design BM about major axis for (e ≤ D/6) case of short column 400x600mm with 8 bars of 16mm ϕ at 40mm clear cover placed equally on four sides. Fck = 20MPa, Fy = 500 MPa; Use WSM	CO2	4
Q. 3	Solve Any Two of the following.		
A)	Describe with reason the stress-strain curve of concrete & steel adopted in LSM	CO3	4
B)	Describe the all serviceability criteria's in limit state method of RCC design	CO3	4
C)	Find development length at a cross section 300x550mm effective with 4 bars of 20mm ϕ in tension & 2 bars of 20mm ϕ in compression, compressed under reaction, subjected to factored BM 300 KNm & factored SF 100KN. Fck = 25MPa, Fy = 500 MPa. Use LSM	CO3	4
Q.4	Solve Any Two of the following.		
A)	Find height of 230mm thick brick wall to be constructed on whole simply supported eff. span of 6.0m of beam section having following	CO4	4

	data. b=300 mm, D=600mm, d=550mm, d'=50, bottom=5 bars 20mm ϕ , top=3 bars of 20 ϕ , Fck = 20 MPa, Fy = 500 MPa		
B)	Design a one central roof beam for hall 6.0x9.0m c/c provided with 300 mm thick peripheral wall & beams at 3.0m c/c carrying 120mm thick slab & 1.5 KN/m ² LL. Assume FF load 1.0 KN/m ² . Fck = 20 MPa, Fy = 500 MPa. Sketch rein. details	CO4	4
C)	Design a slab for room 3.8 m x 6 m having one balcony of 1.2m width on longer side, carrying LL 2.5 KN/m ² and floor finish load of 1KN/m ² . Wall thickness is 300 mm. Fck = 20 MPa, Fy = 500 MPa. Use load factor 1.5 Sketch rein. details	CO4	4
Q. 5	Solve Any Two of the following.		
A)	Design a dog-legged staircase provided in a room 3.0x6.0m having floor to floor height 3.6m. width of stair is 1.4m with well 0.2m. LL is 2.5KN/m ² . Fck = 25MPa, Fy = 500 MPa. Apply load factor 1.5. Sketch rein. details.	CO4	4
B)	Design uniaxially loaded short column having both ends hinged about major axis and top - hinged & bottom- fixed about minor axito carry axial load of 1500 KN and bending moment of 150 KNm about major axis and 80 KNm about minor axis. Use rectangular c/s with 1.5 aspect ratio. Unsupported lengths w.r.t. major and minor axes are 5.0 m and 6.5 m respectively. Consult interaction curves from SP16. Fck=20MPa, Fy=500 MPa, $\gamma=1.5$.Sketch rein. details.	CO5	4
C)	Design rectangular spread footing for column 600 x 800 mm carrying axial load of 1200KN. SBC of soil is 120 KN/m ² , $\phi=30^\circ$ Fck=20MPa, Fy=415 MPa, $\gamma=1.5$.Sketch rein. details.	CO5	4
	*** End ***		